

# **SIMATS ENGINEERING**

## **ASSESSMENT TOOL 3**

**COURSE NAME: DEEP LEARNING**

**COURSE CODE: MLA0408**

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### **COURSE OUTCOME COVERED**

CO2: Analyze Machine Learning and Deep Learning models by evaluating neural network architectures, representation learning, activation functions, unsupervised learning techniques, and their real-world applications. (L4)

CO Weightage: 15%

Assessment Tool Weightage: 20%

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### **QUESTIONS:**

**Total Marks: 20**

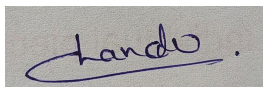
1. Differentiate between Machine Learning and Deep Learning with respect to architecture, data requirements, feature extraction and applications. Give two real-world applications of each.
2. Explain the concept of Representation Learning. Discuss how the width and depth of a neural network influence its learning capability, computational complexity and model performance.
3. Compare ReLU, Leaky ReLU (LReLU) and ELU (Exponential Linear Unit) activation functions. Explain their mathematical behavior, advantages, disadvantages and suitable use cases.
4. Explain the concept of unsupervised learning in neural networks. Describe the working principle of Restricted Boltzmann Machines (RBMs) and Autoencoders and compare their objectives and applications.

**Code of Conduct:**

*I M. Phanith Chandu(192425238) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*

*I understand that any violation of academic integrity rules will result in disciplinary action.*

**Signature of the student:**



**RUBRICS FOR CALCULATION:**

| Criteria                          | Weightage | Level 4 – Exemplary                                            | Level 3 – Proficient                    | Level 2 – Developing                 | Level 1 – Beginning                           |
|-----------------------------------|-----------|----------------------------------------------------------------|-----------------------------------------|--------------------------------------|-----------------------------------------------|
| <b>Concept Understanding</b>      | 25%       | Clearly explains all concepts with complete accuracy and depth | Minor gaps in understanding             | Basic understanding with some errors | Incorrect or very limited understanding       |
| <b>Content Coverage</b>           | 25%       | Covers all required topics comprehensively                     | Covers most topics with minor omissions | Covers some topics only              | Very few topics covered                       |
| <b>Technical Accuracy</b>         | 20%       | All explanations are technically correct and precise           | Minor technical errors                  | Some incorrect or vague explanations | Major technical errors                        |
| <b>Integration of Concepts</b>    | 15%       | Effectively connects and relates concepts logically            | Some connections made with minor gaps   | Limited connections between concepts | No clear relationship between concepts        |
| <b>Clarity &amp; Presentation</b> | 15%       | Well-structured, clear, and easy to understand                 | Mostly clear with minor issues          | Somewhat unclear or poorly organized | Poorly structured and difficult to understand |

## (1) Machine Learning

(1) Uses traditional algorithms such as Decision Trees, SVMs, Random Forest and Linear Regression.

(2) Can produce good results with small to medium-sized datasets.

(3) Features are manually selected and engineered by humans before training the models.

(4) Suitable for structured data such as customer needs, sales data, and financial data.

## Deep Learning

(1) Uses Artificial Neural Networks with multiple hidden layers. Deep Neural Networks.

(2) Requires very large amounts of data to achieve high accuracy.

(3) Automatically learns and extracts important features from raw data without manual intervention.

(4) Best suited for unstructured data such as images, videos, audio, and natural language.

## Real-world Applications :-

### (i) Machine Learning :-

\* Email spam filtering - ML algorithms classify incoming emails as spam or legitimate based on patterns in the email content.

• The system extracts features from emails - word frequencies, presence of suspicious links, sender reputation, header anomalies, and historical user actions. These hand-crafted or automatically selected features feed into a classifier that outputs a spam probability.

\* Credit Risk Scoring / loan approval systems - Faster decisions, reduced human-bias in some cases, better risk pricing, and ability to serve thin-file or under banked customers using various data sources.

## 8.11 Deep Learning:

\* Face Recognition Systems - Used in smartphones, security systems, and attendance systems to identify individual accurately.

\* Autonomous Cars - Deep learning enables vehicles to recognize roads, traffic signs, pedestrians, and other vehicles for safe navigation.

(2) Representation Learning is a technique in Deep Learning where the model automatically learns useful features from raw data instead of relying on manually designed features. It identifies patterns and creates meaningful representations of the input data that help improve prediction accuracy. This eliminates the need for extensive feature engineering and allows the model to handle complex data such as images, text, and audio more effectively.

The width of a neural network refers to no. of neurons in each hidden layer, while the depth refers to the no. of hidden layers in the network. Both play an important role in determining the network's performance.

A wider neural network has more neurons in each layer, enabling it to learn a larger variety of features at the same level. This can improve accuracy for simpler tasks but also increases the no. of parameters, leading to higher memory usage and computational cost. Extremely wide networks may also overfit the training data if regularization techniques are not used.

A deeper neural network contains more hidden layers, allowing it to learn features in hierarchical manner. The first layers learn simple features such as edges or basic patterns, while deeper layers combine them into more complex representations such as objects or faces.

In terms of learning capability, deeper networks are usually more powerful because they can model highly complex relationships in data, whereas wider networks mainly improve the model's capacity to learn more features within a single layer.

Regarding computational complexity, increasing either the width or depth rises the no. of calculations. Wider networks consume more memory due to additional neurons, while deeper networks require more processing time because data must pass through multiple layers during training and inference.

For model performance, the best results are achieved by choosing an appropriate balance between width and depth. A network that is too small may underfit the data, while an excessively large network may overfit and require unnecessary computational resources.



(3)

Activation functions are one of the most important components of Artificial Neural Networks and Deep Learning models. They introduce non-linearity into the network enabling it to learn complex patterns and relationships from data. Without any activation function, a neural network, enabling it to learn complex patterns linear model regard less complex patterns linear model regardless of the number.

Needs for Activation Functions:

i. A neural network behaves like a linear regression model.

ii. Multiple hidden layers cannot learn complex non-linear.

iii. The network cannot classify images, speech, or text efficiently.

iv. The learning capability becomes very limited

Mathematical Formula:

$$f(x) = \begin{cases} 0, & x < 0 \\ x, & x \geq 0 \end{cases}$$

## Working Principle

- \* If the input value is positive, the output is equal to the input.
- \* If the input value is negative, the output becomes zero.

| <u>Input</u> | <u>Output</u> |
|--------------|---------------|
| 0            | 0             |
| -8           | 0             |
| -3           | 0             |
| -1           | 0             |
| 0            | 2             |
| 2            | 5             |
| 5            | 10            |
| 10           |               |

## Advantages :-

### (1) ReLU

- i. Easy to implement and computationally efficient.
- ii. Speeds up model training.
- iii. Reduces vanishing gradient.

### (2) Leaky ReLU

- i. Solves the dying ReLU.
- ii. Performs well in deeper neural networks.

### (3) ELU

- i. Smooth & continuous outputs.
- ii. Fast convergence.

## Disadvantages :-

### (1) ReLU

- i. Suffers from dying ReLU.
- ii. Cannot learn from negative inputs.

### (2) Leaky ReLU

- i. Slightly more complex than ReLU.
- ii. Requires selecting an appropriate  $\alpha$ .

### (3) ELU

- i. Computationally slower.
- ii. Training time is higher.



- (4) Unsupervised learning is a type of machine learning in which the neural network learns patterns from unlabeled data. Instead, the model ~~intend~~ identifies hidden structures, relationships, and patterns within the data.

### Restricted Boltzmann Machine (RBM) :-

A Restricted Boltzmann Machine is a generative neural network that learns the probability distribution of the input data. It consists of two layers.:

i. Visible layer :- Receives input data.

ii. Hidden layer :- Learns hidden features from the data.

There are no connections between neurons within the same layer. Each visible neuron is connected to every hidden neuron. During training, the RBM learns by adjusting the weights between visible and hidden layers using an algorithm called Contrastive Divergence.

### Applications :-

i Feature Extraction

ii Recommendation Systems

iii Collaborative Filtering

iv Dimensionality Reduction

v ~~Deep~~ <sup>train.</sup> Pre-training Deep Neural Networks.

## Autoencoder

An autoencoder is an unsupervised neural network designed to learn an efficient representation of input data by compressing and reconstructing it. It consists of mainly three parts:

i) Encoder - Compresses the input into a lower-dimensional representation.

ii) Latent Space - Stores the most important features of the input.

iii) Reconstructs the original input from the compressed representation - Decoder.

The model is trained by minimizing the difference between the original input and the reconstructed output. This enables the autoencoder to learn meaningful features without requiring labeled data.

## Applications:

i) Image denoising

ii) Feature learning

iii) Image Reconstruction

iv) Data compression

v) Anomaly detection